Product Listing - QA Testing & Bug Fixing

1. Issue #1 - Cart count not showing

- **Location:** `product-list.component.ts` -> `ngOnInit()`
- **Root Cause:** The `cartCount$` observable was declared but never instantiated, leaving the template's `async` pipe listening to an undefined variable.
- **Applied Fix:** Initialized the variable by assigning it to the service's stream: `this.cartCount$ = this.cartService.cartCount$;`.
- **Technical Impact:** Establishes a reactive data stream. The UI now automatically updates whenever the `BehaviorSubject` in the `CartService` emits a new cart count.

2. Issue #2 - Filter doesn't actually filter products

- **Location:** `product.service.ts` -> `filterByCategory()`
- **Root Cause:** The method instantiated a brand-new `BehaviorSubject` on every filter action. This is an anti-pattern that creates disconnected streams and fails to update the component's existing subscription.
- **Applied Fix:** Imported the `of` creation operator from `rxjs` and refactored the method to return `of(filtered)`.
- **Technical Impact:** The `of` operator properly wraps the synchronously filtered array into a lightweight Observable, instantly emitting the correct data to the component without causing memory leaks.

3. Issue #3 - No feedback when adding to cart

- **Location:** `product-list.component.ts` -> `addToCart()`
- **Root Cause:** The function successfully updated the service state, but lacked a visual affordance, leaving users confused as to whether their click was registered.
- **Applied Fix:** Added a native browser confirmation: `alert('${product.name} has been added to your cart!');`.
- **Technical Impact:** Greatly improves User Experience (UX) by providing an immediate, blocking confirmation that the intended action was successful.

4. Issue #4 - Total count doesn't update when filtering

- **Location:** `product-list.component.ts` -> `loadProducts()`
- **Root Cause:** This was a downstream side effect of Issue #2. Because the broken service method failed to pass the filtered data stream to the component, the array length calculation was never triggered.
- **Applied Fix:** Resolved automatically by the architectural fix applied to Issue #2.
- **Technical Impact:** With `filterByCategory()` now correctly emitting the filtered data, the `subscribe` block executes properly, allowing `this.totalCount = data.length;` to accurately sync the UI count with the rendered DOM elements.